

Lecture – 15

Data link layer is divided into 2 layers:

- Upper sublayer (Data link control) => dedicated channel
- Lower sublayer (Multiple access control – MAC) => shared channel

Multiple Access Protocol

Random Access Protocols	Controlled Access Protocols	Channelization Protocols
ALOHA	Reservation	FDMA
CSMA	Polling	TDMA
CSMA/CD	Token Passing	CDMA
CSMA/CA		

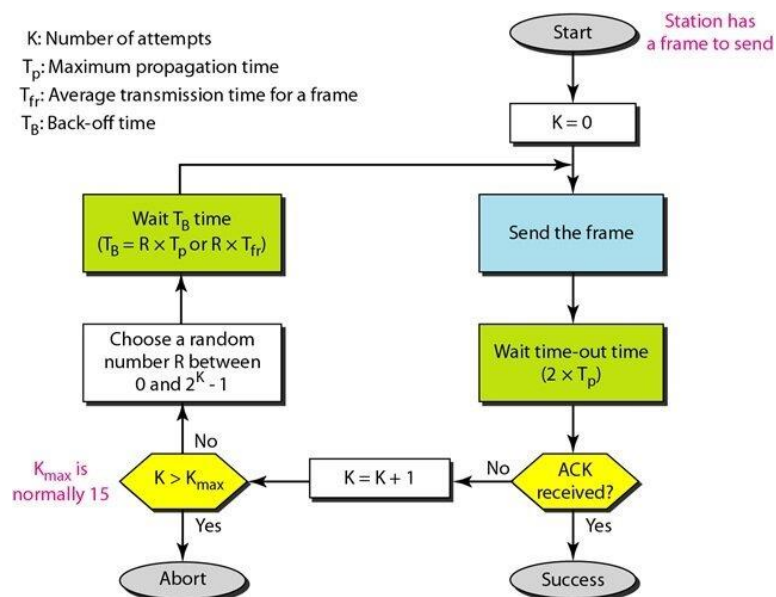
Random Access: Stations compete for channel

Controlled Access: One station controls who transmits

Channelization: divide channel by time, frequency or code

Pure ALOHA

- Transmit immediately when data is ready
- Collisions happen because channel is shared
- Sender waits for ACK, if not received within timeout, assumes collision



Vulnerable Time: period when collision is possible i.e $VT = 2 \times \text{frame transmission time}$

$$\Rightarrow VT = 2T$$

Throughput = $G * e^{-2G}$ $\Rightarrow$ (if G is 0.5, then throughput will be max)

Where, $G \Rightarrow$ average attempts per frame time or total no. of frames offered to channel per frame transmission time

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Slotted Aloha:

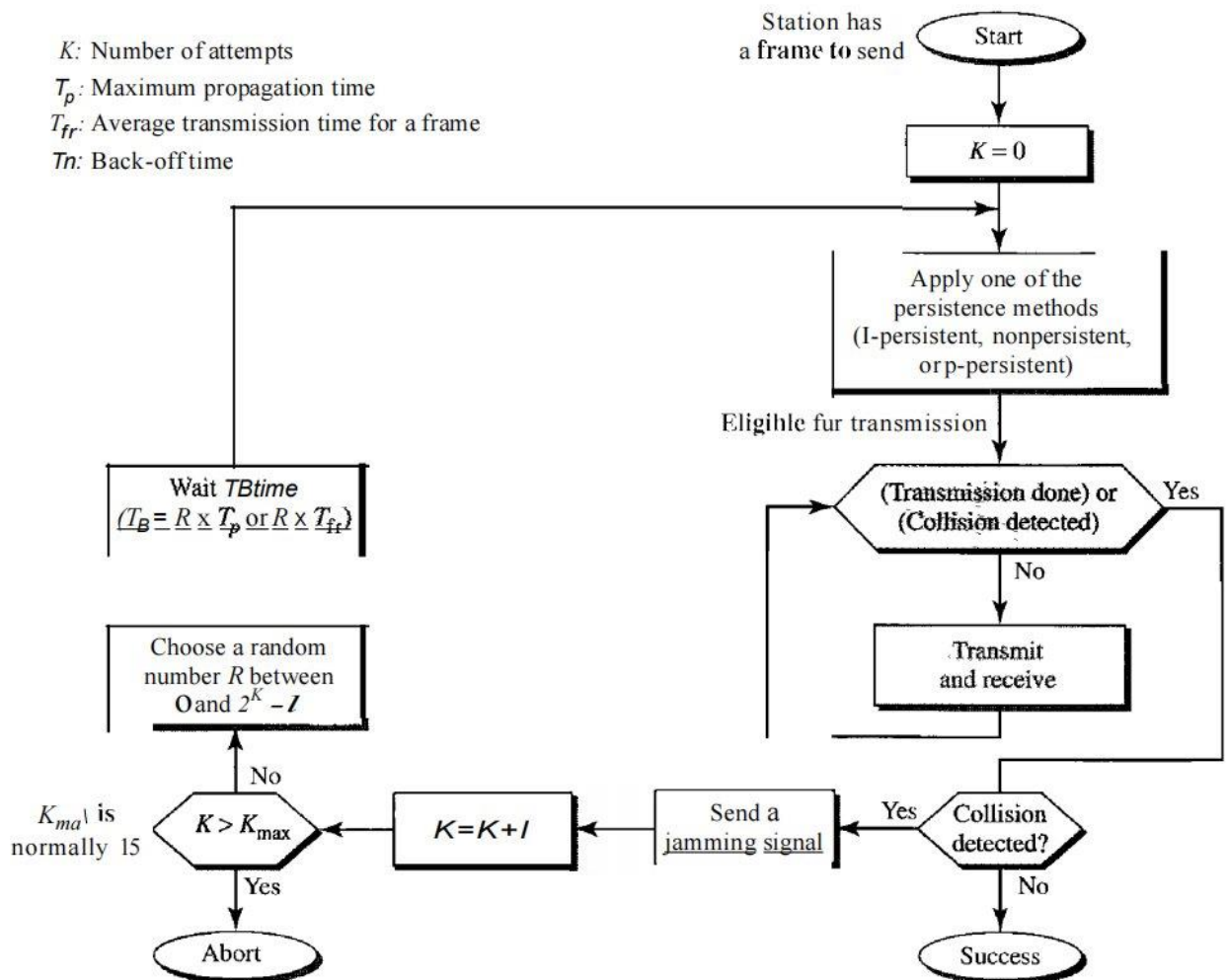
- Vulnerable time = T
- Transmission starts at only synchronized time slots
- Throughput = $G * e^{-G}$ $\Rightarrow$ (if G is 1, then throughput will be max)

CSMA (Carrier Sense Multiple Access):

- Listens before transmitting, transmits only if medium is free but can't eliminate collision completely
- Three types: 1 - persistent, non - persistent, p - persistent.

CSMA/CD (with Collision Detection):

- Listen before transmitting, continue listening while transmitting, then detect collision $\rightarrow$ stop immediately.
- Total time to detect collision: $2T_p$
- To ensure collision detection: $T_{fr} \geq 2T_p$
 $T_{fr} \rightarrow$ Frame transmission time
 $T_p \rightarrow$ Maximum propagation time
- Minimum frame size = $2 \times \text{propagation delay} \times \text{bandwidth}$
- 3 energy levels: zero energy level (idle), normal energy level (single transmission), high energy level (collision).
- Collision occurs when it receives own signal as well as another signal at the same time



- CSMA/CD throughput > ALOHA variants.
- 1-persistent: max ~50% at $G=1$.
- Non-persistent: up to 90% at $G=3-8$. => highest throughput
- P-persistent: ~70–80% (depends on p , G).

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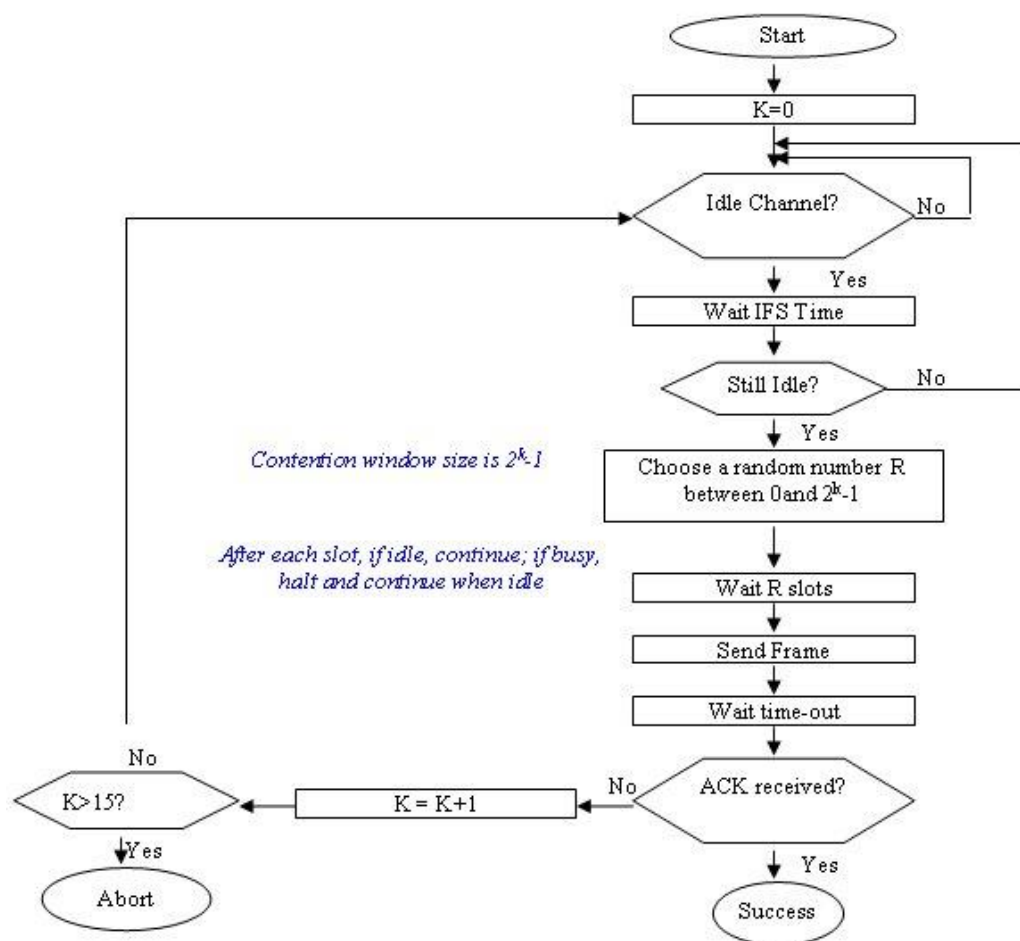
CSMA/CA (Collision Avoidance):

- Wireless signals are weak, collision adds only 5 – 10% extra energy => undetectable
- Wireless stations can't listen and transmit simultaneously

SOLUTION is CSMA/CA

Three Strategies:

- Interframe Space (IFS) - Fixed delay before attempting (less IFS => higher priority)
- Contention Window – Random backoff slots (window doubles if transmission fails)
- Acknowledgements – Confirm successful reception



- Vulnerable time = T_p

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Controlled Access Protocol

Scheduling:

- Scheduling protocols avoid collision.
- *Advantages*: Higher efficiency, lower delay variability, fairness
- *Approaches*: Reservation, Polling, Token passing

Reservation Method:

- Station reserves channel before sending data.
- Reservation system options are:
 - Centralized vs Distributed control:
 - Centralized: One controller collects requests and grants slots
 - Distributed: Stations listen to reservations and complete their own slot.
 - Single vs Multiple frames:
 - Single: One data frame per reservation per cycle.
 - Multiple: Reserve multiple frames at once.
 - Channelized vs Random Access Reservations:
 - Channelized: Fixed dedicated reservation slots (collision – free) & if reservation bit is 1, data goes to transmitting phase.
 - Random accessed: Stations compete for reservation slots.
- Efficiency:
 - Each data frame transmission time = X seconds
 - Each reservation minislot duration = vX seconds
 - The reservation phase is followed by K data frames..

$$\begin{aligned} \text{Total time} &= KX + vX \\ &= (K + v)X \end{aligned}$$

$$\text{Efficiency} = \frac{\text{Useful data transmission time}}{\text{Total cycle time}} \Rightarrow \frac{KX}{(K+V)X} \Rightarrow \frac{K}{K+V}$$

- For Slotted ALOHA, reservation time = $e \cdot vX$

Polling:

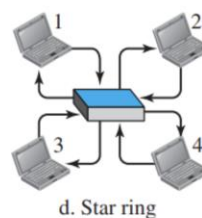
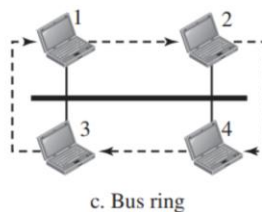
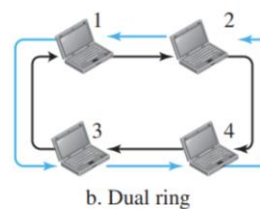
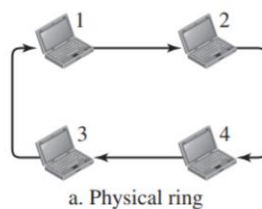
- Primary station controls channel, secondary stations respond.
- Primary polls each secondary sequentially.
- Secondary transmits if it has data, else sends “no data” or it’s idle.
 - Select Function:
 - Primary sends select (SEL) frame to secondary.
 - Secondary acknowledges readiness.
 - Then, primary transmits data.
 - Poll Function:
 - Primary polls secondary to check for outgoing data.
 - Secondary replies with data frame or NAK.
 - Primary sends ACK after successful in data reception.
- Service Limit:
 - Exhaustive: Transmits until buffer empty.
 - Gated: Transmit only frames that arrived before poll.
 - Frame – Limited: One frame per poll.
 - Time – Limited: Transmit upto max time.
- Priority: Place high priority stations at multiple times in polling sequence.

Token passing:

- Stations organized in logical ring => tokens circulates continuously.
- Station with token can transmit, then passes token to successor.
- Wait for token -> capture token, transmit data -> data frame travels to destination -> release token to successor.

Token Passing

Ring Types



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Channelization Protocols: Channelization divides bandwidth among stations.

Three main types:

- FDMA (Frequency Division Multiple Access):
 - Total bandwidth divided into smaller frequency bands.
 - Each user gets a dedicated band.
- TDMA (Time Division Multiple Access):
 - Entire bandwidth is available to all users.
 - Time divided into fixed slots.
 - Each user transmits only in its assigned slot.
- CDMA (Code Division Multiple Access):
 - All users transmit simultaneously on same frequency.
 - Each user has unique spreading code.
 - Receiver uses same code to decode intended signal.
 - Each station has data (d1, d2,) and unique code (c1, c2,).
 - *Orthogonality*: $c_i * c_j = 0$ where $i \neq j$.
 - *Self Correlation*: $c_i * c_i = N$, where N is no.of stations.
 - Mapping => 1 -> +1, 0 -> -1 and idle/silence -> 0.
 - Walsh table:

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} \qquad W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

b. Generation of W_1 , W_2 , and W_4

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CSMA/CD Problems (occurs due to signal range limitations in wireless):

- Hidden Station problem: occurs when two stations cannot hear each other but can communicate with same receiver. => collisions
Example: A -> B <- C (A and C cannot hear each other)
- Exposed Station problem: station unnecessarily stops transmitting because it hears another nearby transmission even if doesn't cause collision.
Example: A <- B C -> D (C hears B but could still transmit)

Network Layer

Services provided:

- Logical Addressing:
- Routing
- Packet forwarding
- Fragmentation & Reassembly
- Congestion Handling
- Inter – networking

Logical Addressing:

- IPv4: 32 bits => 2^{32} addresses (binary notation & dotted decimal notation)
- IPv6: 128 bits => much larger space

Classful Addressing:

- Class A: 0 – 127 (1st byte is netid, 3 bytes hostid)
- Class B: 128 – 191 (1st and 2nd bytes are netid, next 2 bytes are hostid)
- Class C: 192 – 223 (1st, 2nd and 3rd bytes are netid, next byte is hostid)
- Class D: 224 – 239 (multicast)
- Class E: 240 – 255 (reserved)

No. of addresses = $2^n - 2$

All bit zero => network address

All bit ones => broadcast address

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Subnetting:

No. of subnets = $2^n - 2$ (n is no. of 1's)

No. of hosts in subnets = $2^n - 2$ (n is no. of 0's)

Subnet id $\Rightarrow$ and of IP address and subnet mask (host bits) $\Rightarrow$ IP address with the and of IP address and subnet mask

1st subnet id: fix 0's in host bits of subnet mask and then change all the ones to 0 except the last one

Example: subnet mask: 255. 255. 255. 240

240 $\Rightarrow$ 11110000

1st subnet id: 00010000

2nd subnet id: 00100000

last subnet id: 11100000

last host of last subnet id: 11101110

DBA of subnet $\Rightarrow$ Example: 255. 255. 255. 224

224 $\Rightarrow$ 11100000

1st subnet id: 00100000 $\Rightarrow$ DBA: 00111111

4th subnet id: 10000000 $\Rightarrow$ DBA: 10011111

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Subnetting

Fixed Length subnetting:

- All subnets have same size.
- Same subnet mask for all subnets.
- Same no.of hosts per subnet

Variable Length subnetting:

- Subnets can have different sizes.
- Different subnet masks per subnet.
- Also called Classless subnetting.

Categories of IP Address:

- Public IP: Globally unique, Internet accessible (Ex: www.google.com)
- Private IP: local use only, not routable on internet (Ex: devices like computers, printers and mobile in home or office)
- Static IP: Manually configured, permanent. (Ex: public facing like routers, web servers and email services)
- Dynamic IP: Assigned by DHCP (Dynamic Host Configuration Protocol), temporary. (Ex: university network => diff IP address during each session).

Servers use static; clients use dynamic

Casting (delivering data from source to one or multiple destinations):

- Unicast: one-to-one (a single device)
- Broadcast: one-to-all
 - Limited broadcast: all devices on the local network
 - Directed broadcast: all devices on a specific network
- Multicast: one-to-group (a particular group of devices)

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Classless Addressing

Address block restrictions:

- Addresses must be contiguous.
- No. of addresses must be power of 2.
- First address must be divisible by block size.